

AI6124 Assignment 2

Cancer-versus-Non-Cancer Classification with ROC and Equal-Error-Rate Analysis

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Abstract

This report evaluates supervised cancer screening on the NUH ovarian blood-assay data, with the Wisconsin Diagnostic Breast Cancer (WDBC) data as a lower-variance check. Eight representative classifiers are compared using test receiver operating characteristic (ROC) curves and repeated cross-validation. The main NUH target treats borderline tumours as screening-positive together with invasive cancers, because a false negative is the less acceptable error in this screening context; this convention is tested explicitly rather than presented as an unambiguous pathological label. Selecting on training out-of-fold (OOF) scores alone, random forest is chosen (OOF AUC 0.812, OOF equal error rate (EER) 0.236); its OOF threshold of 0.484 then yields test FPR/FNR of 0.214/0.231 on the untouched test split. A pooled repeated cross-validation over all cases, reported separately as a stability summary rather than a held-out estimate, ranks the same model first (AUC 0.845, EER 0.245). Reassigning or excluding borderline cases changes the random-forest CV EER to approximately 0.20, demonstrating material label-policy sensitivity. WDBC results are near saturation (test AUC 0.977–0.997), confirming the implementation on a well-conditioned data set.

1 Task and decision criterion

The task is to learn a binary supervised decision tool, determine the boundary that best balances false positives and false negatives, and plot the ROC curve obtained by varying that boundary. Each model returns a score $s(\mathbf{x})$ and predicts the positive class when $s(\mathbf{x}) \geq \theta$. The scalar *threshold* θ induces the feature-space *decision boundary* $\{\mathbf{x} : s(\mathbf{x}) = \theta\}$.

Sweeping θ traces the ROC curve, with false-positive rate $\text{FPR}(\theta)$ and true-positive rate $\text{TPR}(\theta)$. The false-negative rate is $\text{FNR} = 1 - \text{TPR}$, and the equal error rate is the intersection with $\text{FPR} = \text{FNR}$. Area under the ROC curve (AUC) measures ranking performance independently of a single threshold.

An empirical ROC curve is stepwise. I report an interpolated EER, marked by an asterisk in the tables, but do not imply that its exact intersection is always attainable by one deterministic threshold. The deployable threshold is selected separately from training out-of-fold (OOF) scores as the empirical ROC threshold that minimises $|\text{FPR} - \text{FNR}|$, using the lower balanced error as a tie-breaker. Test-set EERs are descriptive only and are not used for model or threshold selection.

2 Data

2.1 NUH ovarian blood assays

The NUH data accompany the ovarian-cancer study of Tan *et al.* [1]. The supplied group `g2` files contain 28 blood-assay features. Inspection shows that `c1–c4` are one-vs-rest labels of the same patients rather than cross-validation folds: borderline, benign/normal, Stage I–II, and Stage III–IV, respectively. Three borderline test rows duplicate training rows and are removed, leaving 55 training cases and 54 test cases. Across the retained data there are 5 borderline, 56 benign/normal, 14 Stage I–II, and 34 Stage III–IV cases.

The main screening target is **positive** = **borderline** $\cup$ **Stage I–IV** and **negative** = **benign/normal**, giving 53 positive and 56 negative cases. Borderline ovarian tumours are not synonymous with invasive malignancy; grouping them with the positive class is a conservative screening decision intended to avoid a potentially consequential false negative. Section 5.2 reports how the results change when borderline cases are excluded or assigned to the negative class.

Feature identities cannot be mapped reliably because the file column order does not match the source paper; they are therefore denoted $f_1, \dots, f_{28}$. On the training set, f_{23} , f_{24} and f_{28} have maximum-to-median ratios above 20. These columns are transformed by $\log(1 + x)$ before standardisation; the transformation and scaler are fitted inside each training fold.

2.2 Wisconsin Diagnostic Breast Cancer

WDBC contains 569 fine-needle aspirates with 30 nuclear-morphology features, comprising 212 malignant and 357 benign cases [2, 3]. A stratified 80/20 split with seed 0 gives 455 training cases and 114 test cases. The data are close to linearly separable and are used as a sanity check rather than a second clinical claim.

3 Models

Eleven models were implemented and run under the identical protocol; the eight reported here are a representative subset spanning the linear, kernel, ensemble, neural, fuzzy, and pre-trained families. The three omitted models (an extreme learning machine, a predictive-coding network, and a GA-trained MLP) are retained in the accompanying code and result files, and including them does not change the selected model on either data set.

Logistic regression (L_2 , $C = 1$), an RBF-SVM, and a 500-tree random forest provide linear, kernel, and ensemble baselines. The SVM selects $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$ by inner five-fold AUC.

A d -64-32-1 multilayer perceptron (MLP) with tanh hidden units is trained for 300 full-batch epochs using Adam (learning rate 10^{-3} , weight decay 10^{-3}). Two alternatives retain comparable hidden widths while changing the learning mechanism: Forward-Forward uses label-embedded positive and negative examples with layer-local goodness objectives [4], whereas separable CMA-ES optimises the MLP parameters without gradients (population 32, 300 generations) [5]. These methods are retained as compact course-relevant comparisons; no statistical non-inferiority claim is made from their point estimates.

The fuzzy model is a four-rule first-order Takagi–Sugeno ANFIS with Gaussian membership functions and scatter-partition initialisation [6]. TabPFN v3 (package version 8.4.0) [8] is included as a strong pre-trained tabular reference; it performs in-context prediction without task-specific gradient training, following the prior-fitted-network formulation introduced by Hollmann *et al.* [7]. It should not be interpreted as a performance ceiling, and its licence-gated checkpoint is less reproducible than the locally trained models.

4 Evaluation protocol

Every decision is made on training data, and the test split is used once, only to report. Concretely, for the given NUH and WDBC splits, preprocessing is fitted on training data only, and five-fold CV *within the training set* produces OOF scores. Both the choice of model and the choice of threshold are made from those training-OOF scores. The model is then refitted on the complete training set and evaluated once on the test set; the test AUC, interpolated test EER, and the operating point of the selected model are reported afterwards and feed back into nothing.

A separate, purely descriptive summary is also reported. A 54-case test split makes any single estimate volatile, so all cases are additionally evaluated by $5\times$ repeated stratified five-fold CV (25 fits per model), with the preprocessing refitted inside each fold. These pooled CV columns are

computed over the whole cohort and therefore *include the test cases*; they are a resampling summary of how stable a model is across partitions of the available data, not a held-out estimate, and they are not used to select the model, the threshold, or the label policy. Where the pooled ranking is quoted below it is descriptive only, and the selection it would imply is stated separately.

All experiments use seed 0 on CPU. The reproduced environment is Python 3.13.5, scikit-learn 1.9.0, PyTorch 2.13.0, *cma* 4.4.4, and TabPFN 8.4.0.

5 Results

5.1 NUH main screening analysis

Table 1 orders models by the pooled repeated-CV results; Figure 1 shows the test ROC curves. Random forest has the highest pooled CV AUC (0.845 ± 0.066) and the lowest pooled CV EER (0.245 ± 0.089), followed by TabPFN on both metrics. As stated in Section 4, these pooled columns include the test cases and are reported to show stability, not to choose a model; the fold-wise spreads are in any case larger than most differences between methods (Figure 2), so the ordering is descriptive.

Table 1: NUH-g2 main screening label (borderline positive). Test EER* is the interpolated ROC intersection. CV columns are 5×5 repeated-CV over all 109 cases (mean $\pm$ fold-wise s.d.); because they pool the training and test cases they are a stability summary, not a held-out estimate, and the model was selected on training-OOF scores instead (Section 4).

Model	Test AUC	Test EER*	CV AUC	CV EER
LogReg	0.691	0.346	0.773 ± 0.083	0.272 ± 0.083
SVM-RBF	0.805	0.250	0.809 ± 0.085	0.282 ± 0.091
RandomForest	0.789	0.231	0.845 ± 0.066	0.245 ± 0.089
MLP-BP	0.640	0.357	0.764 ± 0.083	0.286 ± 0.074
Forward-Forward	0.820	0.269	0.813 ± 0.082	0.272 ± 0.090
CMA-ES-MLP	0.746	0.231	0.801 ± 0.073	0.269 ± 0.063
ANFIS	0.673	0.385	0.749 ± 0.121	0.302 ± 0.108
TabPFN	0.765	0.250	0.828 ± 0.064	0.255 ± 0.076

NUH-g2: test-set ROC curves; dots mark the interpolated equal-error point (dashed: FPR = FNR)

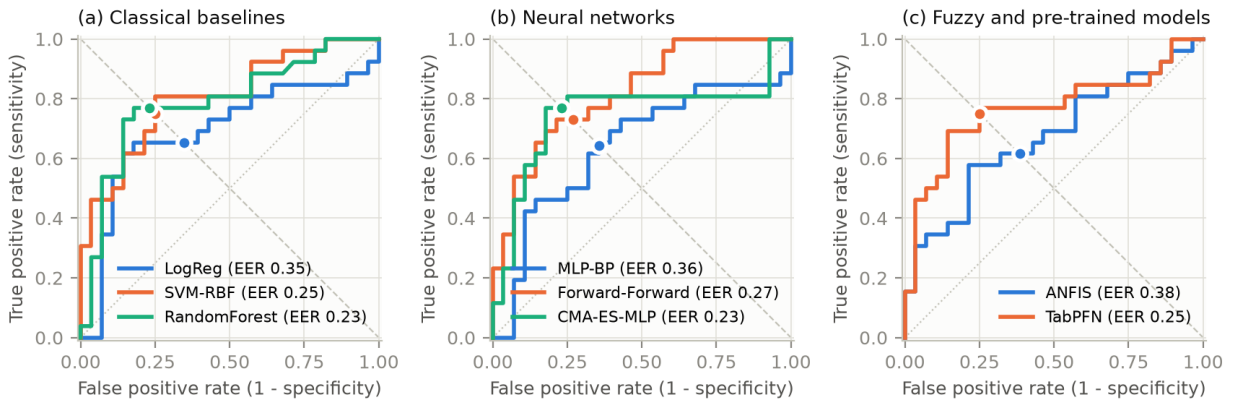


Figure 1: NUH test ROC curves for the reported model set, one panel per model family. Dots mark linearly interpolated EER points; a dot can lie between two empirical operating points and need not correspond to one deterministic threshold.

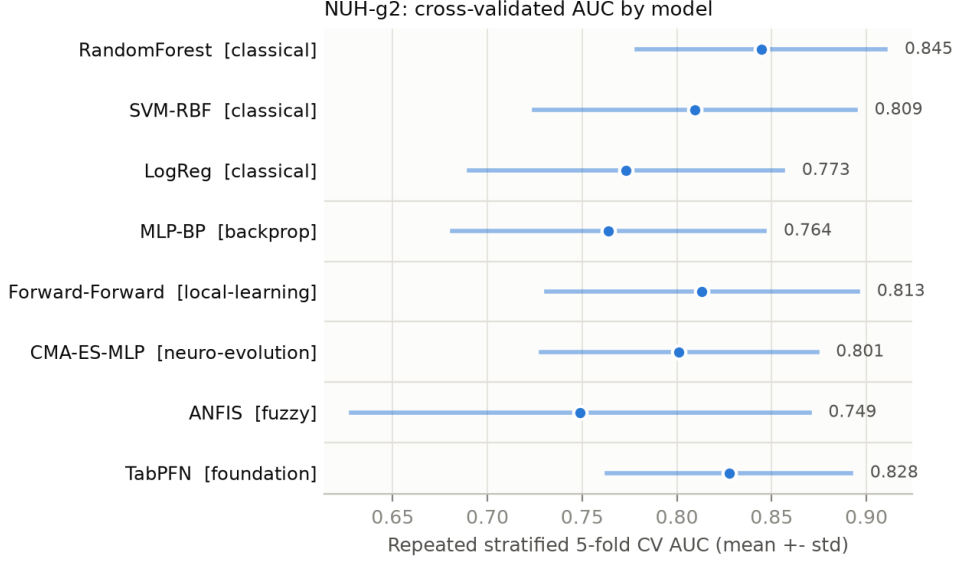


Figure 2: NUH pooled repeated-CV AUC over all 109 cases (mean $\pm$ fold-wise s.d. over 25 fits), grouped by model family. The intervals overlap substantially, so the ordering in Table 1 is descriptive rather than evidence of a reliable difference between methods; as in Table 1, these values pool training and test cases and were not used for selection.

Selected decision boundary. Model selection uses the training-OOF scores alone: neither the test set nor the pooled CV columns take part. On the training OOF scores random forest is best on both criteria (AUC 0.812, EER 0.236; next best TabPFN, 0.796 and 0.250), so it is selected, and its closest attainable training-OOF equal-error threshold is $\theta_{\text{OOF}} = 0.484$ on the vote fraction. Applied after refitting, this threshold gives test FPR 0.214, FNR 0.231, sensitivity 0.769, specificity 0.786, and accuracy 0.778; the interpolated test EER is 0.231. CMA-ES-MLP happens to reach the same descriptive test EER, but it was not selected: its training-OOF EER is clearly worse (0.296), and the test-set values were not available to the selection step. That the pooled CV ranking in Table 1 also puts random forest first is a consistency check reported after the fact, not the reason for the choice.

5.2 Sensitivity to the borderline label

The source data distinguish borderline tumours from both benign/normal cases and invasive cancer. I therefore repeat the pooled 5×5 CV for SVM, random forest, and MLP under three policies: borderline as screening-positive (main analysis), excluded, and assigned to non-cancer. This comparison is deliberately run on the pooled cohort, because the question is how the *target definition* changes the achievable error rate rather than how one fitted model generalises; for the same reason the numbers below are not held-out estimates and none of them was used to choose the reported model or its threshold.

The policy materially changes the point estimates. For random forest, excluding borderline cases changes CV AUC/EER from 0.845/0.245 to 0.882/0.195; assigning them to non-cancer gives 0.885/0.196. SVM EER changes from 0.282 to 0.250 or 0.243, while MLP changes from 0.286 to 0.282 or 0.256. These differences are consistent with borderline cases being difficult to distinguish from benign cases [1], but the fold-wise spreads overlap and the sample contains only five borderline patients. The experiment demonstrates sensitivity to the target definition; it does not establish that labelling borderline cases as non-cancer is clinically preferable. The main screening-positive policy is retained because it is the more conservative choice when false negatives are considered more serious.

Table 2: NUH borderline-label sensitivity. Counts are retained total cases with cancer/non-cancer counts in parentheses.

Borderline policy	Model	n (cancer/non-cancer)	CV AUC	CV EER
Cancer	SVM-RBF	109 (53/56)	0.809 ± 0.085	0.282 ± 0.091
Cancer	RandomForest	109 (53/56)	0.845 ± 0.066	0.245 ± 0.089
Cancer	MLP-BP	109 (53/56)	0.764 ± 0.083	0.286 ± 0.074
Excluded	SVM-RBF	104 (48/56)	0.828 ± 0.078	0.250 ± 0.095
Excluded	RandomForest	104 (48/56)	0.882 ± 0.065	0.195 ± 0.082
Excluded	MLP-BP	104 (48/56)	0.762 ± 0.091	0.282 ± 0.092
Non-cancer	SVM-RBF	109 (48/61)	0.832 ± 0.086	0.243 ± 0.077
Non-cancer	RandomForest	109 (48/61)	0.885 ± 0.066	0.196 ± 0.082
Non-cancer	MLP-BP	109 (48/61)	0.779 ± 0.104	0.256 ± 0.120

5.3 WDBC sanity check

All reported models perform strongly on WDBC (Table 3): test AUC ranges from 0.977 to 0.997, rather than being strictly above 0.98 for every model. TabPFN has the best test AUC (0.997), interpolated test EER (0.024), pooled CV AUC (0.997 ± 0.004), and pooled CV EER (0.023 ± 0.019). WDBC is used only as an implementation check, so no model is selected on it and these figures are reported descriptively; the near-saturated results confirm that the ROC/EER pipeline behaves as expected on a larger, well-conditioned data set.

Table 3: WDBC results. Definitions are the same as in Table 1.

Model	Test AUC	Test EER*	CV AUC	CV EER
LogReg	0.993	0.048	0.995 ± 0.005	0.029 ± 0.020
SVM-RBF	0.994	0.048	0.996 ± 0.006	0.028 ± 0.022
RandomForest	0.977	0.048	0.990 ± 0.009	0.041 ± 0.021
MLP-BP	0.989	0.071	0.995 ± 0.006	0.032 ± 0.019
Forward-Forward	0.988	0.069	0.994 ± 0.008	0.028 ± 0.017
CMA-ES-MLP	0.984	0.069	0.994 ± 0.008	0.032 ± 0.020
ANFIS	0.992	0.048	0.994 ± 0.006	0.030 ± 0.019
TabPFN	0.997	0.024	0.997 ± 0.004	0.023 ± 0.019

6 Discussion

The assignment requirements are met directly: the classifiers return continuous scores, the ROC curves are obtained by sweeping the threshold, and the selected random-forest boundary is determined from training OOF data. Separating the interpolated EER from the attainable threshold avoids claiming an exact FPR–FNR equality that a finite test set cannot necessarily realise.

The NUH results are dominated by small-sample uncertainty and label definition. The training-OOF comparison supports random forest as the preferred choice among the tested models under the stated protocol, and the pooled repeated CV agrees; neither shows that alternative learning rules are equivalent or non-inferior, and the pooled figures cannot be read as out-of-sample performance because they include the test cases. Forward-Forward and CMA-ES have competitive point estimates, while ANFIS is more variable; stronger claims would require paired statistical analysis across repeated splits and a larger external cohort.

The EER objective gives false positives and false negatives equal weight, as required by the task. A clinical screening threshold would normally favour sensitivity and must be chosen using explicit costs, prevalence, and clinical validation. Treating borderline cases as screening-positive addresses the label side of that concern but does not replace cost-sensitive threshold selection.

Limitations include the five borderline cases, the reverse-engineered NUH file structure, fixed hyperparameters for most models, and reliance on one local version stack. TabPFN additionally

requires licensed weights. Exact dependency versions and data checksums should be retained with the submitted code because an unpinned package list alone is insufficient for long-term reproduction.

7 Conclusion

Random forest is the preferred decision tool for the main NUH screening definition. It was selected on training-OOF scores alone (OOF AUC 0.812, OOF EER 0.236), and its training-OOF vote threshold of 0.484 yields 0.769 sensitivity and 0.786 specificity on the test split, which was held out from both the model choice and the threshold choice. The pooled repeated CV, which ranks the same model first (AUC 0.845, EER 0.245), is reported only as a stability check and is not an out-of-sample estimate. The ROC analysis satisfies the requested decision-boundary sweep, while the WDBC results provide a successful implementation check. Most importantly, the borderline sensitivity experiment shows that NUH performance estimates depend materially on whether five clinically ambiguous cases are treated as positive, excluded, or treated as negative. The reported result is therefore conditional on the explicitly stated conservative screening label.

8 AI Usage Disclosure

This assignment was written by the student, with the assistance of AI tools (ChatGPT and Claude) for grammar, style checking and code generation. The student has verified that all content is original and that any AI-generated suggestions have been critically evaluated and appropriately integrated.

References

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